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Batch - 4.

## LSTM Practice questions

**Ques 1.** Input $x_t = 0.5$
Prev. hidden state $h_{t-1} = 0.2$
Prev. cell state $C_{t-1} = 0.4$

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

$$\tanh(z) = \frac{e^{z} - e^{-z}}{e^{z} + e^{-z}}$$

i) forget gate : $f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f)$

Substitute values :

$$f_t = \sigma(0.7(0.5) + 0.3(0.2) + 0.1)$$
$$= \sigma(0.0510)$$

Sigmoid : $\frac{1}{1+e^{-0.0510}}$

$\approx 0.6248$

$$\boxed{f_t = 0.6248}$$

The forget gate retains approximately 62.48% of the prev. cell state information.

ii) input gate : $i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i)$

$i_t = \sigma(0.6(0.5) + 0.4(0.2) + 0.0)$

$i_t = \sigma(0.3800)$

Sigmoid : $\frac{1}{1+e^{-0.3800}}$

$$\boxed{i_t = 0.5938}$$

The input gate controls how much new candidate information enters the cell state.

iii) Candidate state : $\tilde{C}_t = \tanh(W_c x_t + U_c h_{t-1} + b_c)$

Substitute values :

$\tilde{C}_t = \tanh(0.9(0.5) + 0.2(0.2) + 0.0)$

$\tilde{C}_t = \tanh(0.4900)$

$\tilde{C}_t = \frac{e^{0.4900} - e^{-0.4900}}{e^{0.4900} + e^{-0.4900}}$

$$\boxed{\tilde{C}_t = 0.4542}$$

iv) New cell state $C_t$:

$$C_t = f_t C_{t-1} + i_t \tilde{C}_t$$

$$C_t = (0.6248)(0.4) + (0.5938)(0.4542)$$

$$\boxed{C_t = 0.5196}$$

The new cell state contains both retained information from the prev. state and new candidate info.

v) Output gate : $O_t$

$$O_t = \sigma(W_o x_t + U_o h_{t-1} + b_o)$$

$$O_t = \sigma(0.4)$$

$$O_t = \frac{1}{1+e^{-0.4}}$$

$$\boxed{O_t = 0.6100}$$

vi) Hidden state $h_t$

$$h_t = O_t \tanh(C_t)$$

$$h_t = 0.6100 \tanh(0.5196)$$

$$h_t = 0.2912$$

Ques 2. Input Dim. $D = 10$
Hidden units $H = 32$

a) One bias vector per gate

For each gate, the trainable parameters are:

1. Input weight : $D \times H$
2. Recurrent weight : $H \times H$
3. Bias : $H$

for 1 gate : $P_{gate} = DH + H^2 + H$

$P_{gate} = (10)(32) + (32)(32) + 32$.

$P_{gate} = 1376$

For 4 gates : $P_{total} = 4 P_{gate}$.

$= 4 (1376)$

$$P_{total} = 5504$$

b) Two bias vector per gate:

$P_{gate} = DH + H^2 + 2H$

$P_{gate} = 320 + 1024 + 64$
$= 1408$

For 4 gates: $P_{total} = 4(1408)$

$$\boxed{P_{total} = 5632}$$

Ques 3. The gradient scaling factor is approximated by:

$$\text{Scaling factor} = f^K$$

$f$: constant forget gate value
$K$: no. of time steps.

a) $f = 0.9$

After 10 steps:

$$S.F. = 0.9^{10}$$
$$= 0.3486784401$$

Rounded to 4 decimal places:

$$\boxed{0.3487}$$

The gradient retains approx. 34.87% of its original magnitude.

After 50 steps:

$$S.F = 0.9^{50}$$
$$\approx 0.0051538$$
$$= 0.0052$$

The gradient retains approx 0.52% of its original magnitude.

(b) $K = 50$

Case 1: $f = 0.5$

$$S.F = 0.5^{50}$$
$$= 8.8818 \times 10^{-16}$$
$$= 0.0000$$

The result is zero, although the actual value is extremely small and not exactly zero.

Case 2: $f = 0.99$

$$S.F = 0.99^{50}$$
$$\approx 0.6050$$

The gradient retains approx. 60.50% of its original magnitude.

(c) Interpretation

When the forget gate is 0.5 or 0.9, the gradient becomes very small over 50 time steps, making it difficult for the network to learn long term dependencies.
When the forget gate is 0.99, the gradient retains about 60.50% of its magnitude.
LSTM's help reduced vanishing gradients through their cell state and learned forget gate which can preserve information over long sequences: they do not eliminate the problem in every situation.

Ques4. (a) Total hourly records.

One day contains 24 hours.

One year: $365 \times 24 = 8760$ hrs.

Three years: $N = 3 \times 8760$

$N = 26,280$ records

(b) no. of input/output windows.

Each window requires:
168 input records
24 subsequent target records.

with a stride of 1 hr, the no. of valid windows is:

Window = N - lookback - horizon + 1
= 26280 - 168 - 24 + 1
= 26089

The +1 accounts for the just valid window.

(c) Tensor shapes for a batch of 64 windows.

The standard input shape is:
(batch size, time steps, features)

X = (64, 168, 3)

| X : (64, 168, 3) |

For 24 hr forecast, assuming the model predicts one load value per hour:

| y : (64, 24) |

This means:
- 64 separate windows
- each ip. contains 168 hourly records and 3 features per record.
- each op. contains 24 predicted hourly load values.

(d) Training window and num. batches.

The first 70% of records are used for training.

Step 1: Calculate training records.

$$N_{train} = 0.70 \times 26280$$
$$= 18,396$$

Step 2: Calculate training windows

$$W_{train} = N_{train} - 168 - 24 + 1$$
$$= 18,205$$

Step 3: Calculate num. batches.

$$\text{Batches} = \left\lceil \frac{\text{Training window}}{\text{Batch size}} \right\rceil$$

$$= \left\lceil \frac{18205}{64} \right\rceil$$

Since: $64 \times 284 = 18176$

Remaining windows: $18205 - 18176 = 29$.

There are 284 full batches and one final partial batch.

285 mini-batches per epoch.

(e) LSTM parameter count:

Given: input dim $D = 3$
hidden size $H = 64$
One bias vector per gate.

$$P = 4H(D + H + 1)$$

$$P = 4(64)(3 + 64 + 1)$$

$$\boxed{P = 17,408}$$ trainable parameters

Ques5. (a) Min - Max normalization

Given: min. reading = 20 mm/s
max. " = 120 mm/s
actual " = 95 mm/s

$$x_{norm} = \frac{x - x_{min}}{x_{max} - x_{min}}$$

$$= \frac{95 - 20}{120 - 20}$$

$$\boxed{x_{norm} = 0.7500}$$

normalised reading is 0.75, lies in range 0 and 1.0

(b) failure prediction using threshold.

Given:
Predicted failure prob = 0.83
decision threshold = 0.5

Decision rule:

$$\therefore \hat{y} = \begin{cases} 1, & p \geq 0.5 \\ 0, & p < 0.5 \end{cases}$$

Compare

$$0.83 \geq 0.5$$

∴ **Prediction = 1 (failure)**

The model predicts that the pump will fail within the next cycle.

(c) Confusion matrix and evaluation matrix:

Given: total cycles = 1000
actual failures = 40
predicted " = 45
correctly " = 30

→ **TP = 30**

FP = Pred. failure − TP
= 45 − 30

→ **FP = 15**

→ FN = actual failure − TP
= 40 − 30

**FN = 10**

→ TN = 1000 − 40 − 15

**TN = 945**

Confusion matrix :

| actual / predicted | failure | no failure | total |
|---|---|---|---|
| failure | TP = 30 | FN = 10 | 40 |
| no failure | FP = 15 | TN = 945 | 960 |
| total | 45 | 955 | 1000 |

$$\text{Precision} = \frac{TP}{TP+FP}$$

$$= \frac{30}{30+15} = \frac{30}{45}$$

$$= 66.67\% \mid 0.6667$$

$$\text{Recall} = \frac{TP}{TP+FN}$$

$$= \frac{30}{30+10}$$

$$= \frac{30}{40}$$

$$= 0.7500 \mid 75\%.$$

$$\text{F1-score} = \frac{2PR}{P+R}$$

$$= \frac{2(30/45)(30/40)}{(30/45)+(30/40)}$$

$$F_1 = \frac{60}{85}$$

$$= 0.7059 \;/\; 70.59\%$$

$$\text{accuracy} = \frac{TP + TN}{N}$$

$$= \frac{30 + 945}{1000}$$

$$= \frac{975}{1000}$$

$$= 97.50\%$$

(d) The maintenance team should generally prioritize recall, because missing a genuine pump failure can lead to equipment damage, costly downtime, or safety risks.

Recall is 75% which means the model detects 30 out 40 actual failure and misses 10.

Accuracy can be misleading because only 40 out of 1000 cycles are followed by a failure. A model that predicts no failure for every cycle would achieve:

$$\text{accuracy} = \frac{960}{1000} \times 100$$
$$= 96\%$$

Yet it would detect zero actual failures giving a recall of 0%.

∴ accuracy alone is not enough. The team should evaluate recall alongside precision, recall, F1-score, and the operational cost of false alarm.